

MAT9004 Assignment 1

Due: Friday, 16 August 2024, 23:55 PM (End of Week 4)

Please write up your solutions neatly! In particular, use notation and terminology correctly and include some necessary comments. Partial marks will be given for showing that you know some aspects of the answer, even if your solution is incomplete. The questions are NOT arranged in order of difficulty, so you should attempt every question.

Both handwritten and typed solutions are accepted. Please make sure you convert all your work into **on single pdf file** as other types might not display properly on moodle.



1. [10 points] **Product Pricing:** A manufacturer of nails notes that most customers buy a single pack of nails at a time, but will buy a bigger pack if it is cheaper per nail. A marketing company was engaged to conduct a survey and found the following prices that customers were willing to pay for packs of nails of different sizes:

n.nails:	50	100	200	300	500	750	1000
price (\$):	2.75	3.50	4.45	5.15	6.15	7.10	7.85

For ease of use with Python, Matlab, etc here is the same data as arrays (quantity q_i and price p_i):

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q = [50,100,200,300,500,750,1000]
p = [2.75, 3.5, 4.45, 5.15, 6.15, 7.1, 7.85]
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- (a) [3 points] Of the three basic functions for data analyses discussed in the lectures (power law, exponential and logarithmic), which might apply for this data? Choose which you think is best and provide reasons. Include at least one plot with your answer.
 - (b) [2 points] Fit a line $y = mx + c$ to the data (transformed as necessary) depending on your answer for the previous part. State clearly what the coefficients m and c are that you have calculated. Then determine the original function $p(q)$ giving price as a function of quantity. It is sufficient to provide your answer to two decimal places.
Hint: There is no need to use a least squares line fitting software or the like. You may simply estimate the line based on the two end points, or any other approach you choose. However, please state clearly how you determined the line.
 - (c) [3 points] The manufacturing of nails has a fixed cost of 0.3 cents per nail with a fixed cost of 8 cents for the packet. Determine a function $f(q)$ that gives the profit (difference between price and cost) for packets of different sizes. Then state the derivative of this function. Determine for what packet size the company will make the maximum profit and what that profit is.
 - (d) [2 points] Given the meaning of the function f in this application, what is the domain and the range of $f(q)$?
2. [5 points] Consider the function

$$g(x) = \frac{5}{18}x^3 + \frac{1}{6}x^2 + 5 - \frac{1}{\ln(2)}2^x$$

- (a) [1 point] What is the derivative $g'(x)$?
- (b) [2 points] For what range around $x = 1$ is the function $g(x)$ invertible?
Hint: You will need to find a solution to an equation numerically (should be easy to do by trial-and-error).

- (c) [2 points] Is the function $g(x)$ convex over the interval $[0, 1]$? Use the second derivative of g to justify your answer.
3. [15 points] A regional health authority would like to locate an ambulance station so that the service is as close as possible to all towns to be serviced by this ambulance. There are n towns. All the towns are connected by a major highway. Hence the location of each town $i = 1, \dots, n$ can be defined by a single number d_i giving the distance along the highway from the first town.
- (a) [4 points] The location of the ambulance station x is to be optimised. The objective (penalty to be minimised) is the average *square* of the distance between the station and all towns. (This penalises having towns far from the station more strongly than nearby towns). Write down the objective function $f(x)$ and derive an expression for the optimal x in terms of the d_i for $i = 1, \dots, n$.
- (b) [6 points] Some people on the board of the health authority disagree with the objective function and would like to minimise simply the average distance of the station from all towns. (i) Write down the objective function $g(x)$ that gives the average (non-squared) distance of the station at x to all of the towns $i = 1, \dots, n$. (ii) explain why the minimum of $g(x)$ can *not* be found by solving $g'(x) = 0$, (iii) provide a plot of $g(x)$ if $n = 9$ and the distances are:
 $d = [0, 6, 13, 22, 39, 50, 69, 90, 108]$
 (iv) find the $\min g(x)$ (e.g. by inspection from the plot) and discuss how you can be sure there is no better solution outside the range of the plot, (v) compare this solution (minimum of g) with the optimal solution to $\min f(x)$.
- (c) [5 points] The region decided to purchase a helicopter to more directly service the towns. Now they need to locate a helipad to best service towns. Assume that the aim is to minimise the average squared distance between the helipad at location (x, y) (distance east-west and north-south) with town locations given by coordinates (u_i, v_i) . Since towns have different population sizes (p_i), the optimal helipad location is to be based on minimising the population average of squared distances. That is, the average squared distance per person, or sum of population size times squared distance all divided by the total population. (For this exercise we take the distance to each person to be the distance to the town centre).
- (i) Formulate the objective as a function of both x and y then show that this is really the sum of two independent functions ($f(x) + g(y)$), so that the problem can be solved by minimising $f(x)$ and $g(y)$ independently. (ii) Obtain the derivative of these functions to find the optimal solution x, y in terms of the data $(u_i, v_i$ and p_i parameters). (iii) Show that the problem is convex using the second derivative of the two functions.
- Hint:* To compute the distance between two location (x, y) and (a, b) we can use Pythagoras' theorem: $\text{distance} = \sqrt{(x - a)^2 + (y - b)^2}$.

Total: 30 points